

Variational Quasi-Trefftz Method for Time-Harmonic First-Order Wave Propagation Systems

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Abstract

This work presents a quasi-Trefftz framework aimed at improving the conditioning of classical Trefftz methods. The proposed approach relies on the introduction of an auxiliary solver used to construct basis functions that are significantly less correlated.

Keywords: Helmholtz equation, Quasi-Trefftz, Discontinuous Galerkin, first-order system.

1 Introduction

In the frequency domain, the numerical solution of wave propagation problems using classical discretization methods requires the LU factorization of the matrix appearing on the left-hand side. Unfortunately, this leads to extremely high memory costs due to fill-in in the matrix sparsity pattern, even when reordering strategies are used.

However, domain decomposition strategies can be employed to enable the use of iterative solvers together with matrix-free techniques, which drastically reduce the computational cost.

Variational Trefftz methods can be interpreted as an algebraic reformulation of domain decomposition methods, making them particularly well suited for high-performance computing (HPC). However, when the Galerkin space is constructed from sums of plane waves, it is well known that these methods suffer from severe round-off errors, which significantly deteriorate the quality of the numerical results.

In this talk, we present a unified framework for Trefftz methods together with the definition of a new quasi-Trefftz approximation space that significantly improves the robustness and accuracy of the numerical method.

2 Time-harmonic first-order systems

Symmetric Friedrichs hyperbolic systems provide a unified framework for the modeling of wave propagation phenomena. They encompass

a wide range of physical models, including acoustic waves in fluids, Maxwell's equations in vacuum, and the equations of linear elasticity.

These systems can be written in the form

$$M \partial_t u + \sum_{i=1}^d F_i \partial_{x_i} u = 0 \quad \text{in } \Omega \subset \mathbb{R}^d, \quad (1)$$

where $u : \Omega \times \mathbb{R} \rightarrow \mathbb{R}^m$ is the unknown field, $M \in \mathbb{R}^{m \times m}$ is a symmetric positive definite matrix, and $(F_i)_{1 \leq i \leq d}$ are symmetric matrices of size $m \times m$.

In the time-harmonic regime we seek solutions of the form

$$u(x, t) = \Re(u(x)e^{-i\omega t}),$$

which leads to the time-harmonic first-order system

$$-i\omega Mu + \sum_{i=1}^d F_i \partial_{x_i} u = 0 \quad \text{in } \Omega. \quad (2)$$

Boundary conditions are expressed in terms of the normal flux matrix

$$F(\mathbf{n}) = \sum_{i=1}^d n_i F_i,$$

where $\mathbf{n}$ denotes the unit outward normal to $\partial\Omega$. The matrix $F(\mathbf{n})$ admits the spectral decomposition

$$F(\mathbf{n}) = F^+(\mathbf{n}) + F^-(\mathbf{n}),$$

where $F^+(\mathbf{n})$ and $F^-(\mathbf{n})$ correspond to the positive and negative parts of the spectrum.

Boundary conditions are imposed on the incoming characteristic variables associated with the negative eigenvalues of $F(\mathbf{n})$. For instance, an absorbing boundary condition can be written as

$$F^-(\mathbf{n})u = F^-(\mathbf{n})u_{\text{inc}},$$

where u_{inc} denotes an incident field.

This unified structure turns out to be particularly well suited for Trefftz discretization. In

particular, we observed that it yields an algebraic framework tailored for HPC, enabling the systematic and largely automated design of Trefftz methods.

3 Trefftz variational formulation

Let $\mathcal{T}_h$ be a mesh of Ω and denote by $\mathcal{F}_h$ the skeleton of the mesh. For each element $K \in \mathcal{T}_h$, we introduce the local Trefftz space

$$T(K) = \left\{ v \in (L^2(K))^m : \right. \\ \left. -i\omega Mv + \sum_{i=1}^d F_i \partial_{x_i} v = 0 \quad \text{in } K, \right. \\ \left. F(\mathbf{n})v \in L^2(\partial K) \right\}.$$

The global Trefftz space is then defined as

$$T_h = \{v \in L^2(\Omega)^m : v|_K \in T(K), \forall K \in \mathcal{T}_h\}.$$

Since the functions in T_h satisfy the differential equation inside each element, the volume contributions vanish and the variational formulation involves only interface contributions over the mesh skeleton.

The Trefftz variational formulation reads: find $u_h \in T_h$ such that for all $v_h \in T_h$

$$\sum_{K \in \mathcal{T}_h} \int_{\partial K} (F^+(\mathbf{n}_K) u_K + F^-(\mathbf{n}_K) u_V) \cdot \bar{v}_K ds = 0.$$

Here u_K denotes the restriction of u_h to the element K , while u_V denotes the trace coming from the neighbouring element across the considered interior face. On boundary faces, u_V is replaced by the incident field u_{inc} .

4 Quasi-Trefftz discrete space

Classically, discrete Trefftz spaces are constructed from exact local solutions of the governing equation (plane waves, Bessel functions, evanescent modes). However, this approach may lead to large numerical errors due to ill-conditioning, which in three dimensions may severely deteriorate the quality of the approximation or even cause the numerical method to fail.

Our main idea is to define the functions of the Trefftz space not as exact solutions of the first-order system, but as highly accurate approximations obtained by solving a local first-order system numerically in each element. The numerical solver must be sufficiently accurate

in order to avoid introducing additional consistency errors.

More precisely, for each element K , we consider the local problem

$$-i\omega Mv + \sum_{i=1}^d F_i \partial_{x_i} v = 0 \quad \text{in } K, \quad (3)$$

together with the boundary condition

$$F^-(\mathbf{n})v = \varphi \quad \text{on } \partial K, \quad (4)$$

where φ belongs to a suitable piecewise polynomial space defined on ∂K .

5 Content of the talk

In this talk, we present the construction of the quasi-Trefftz formulation based on a Flux Reconstruction method [2]. This approach is particularly well suited for the type of local problems considered here and enables the construction of highly accurate quasi-Trefftz basis functions.

Moreover, the impact on the conditioning of the method and on the quality of the approximation will be investigated. In particular, we establish a clear relationship between the degree of the polynomial trace space and the degree of the Flux Reconstruction method.

Finally, an optimization of the decomposition of the flux operator $F(\mathbf{n})$ into $F^+(\mathbf{n})$ and $F^-(\mathbf{n})$ using artificial intelligence techniques will be performed in order to further improve the quality of the Trefftz method.

More details and references can be found in the PhD thesis of Matthias Rivet [3].

References

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